

# Cell Wall Synthesis Inhibitors 1

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# Cell Wall Synthesis Inhibitors

Penicillin , cephalosporin, carbapenems, monobactams, lipoglycopeptides, Fosfomycin, polymyxin B, vancomycin, and daptomycin.

A. Mechanism of action

B. Antibacterial spectrum

C. Resistance

D. Pharmacokinetics

✓ Administration:

✓ Absorption:

✓ Distribution:

✓ Metabolism &Excretion:

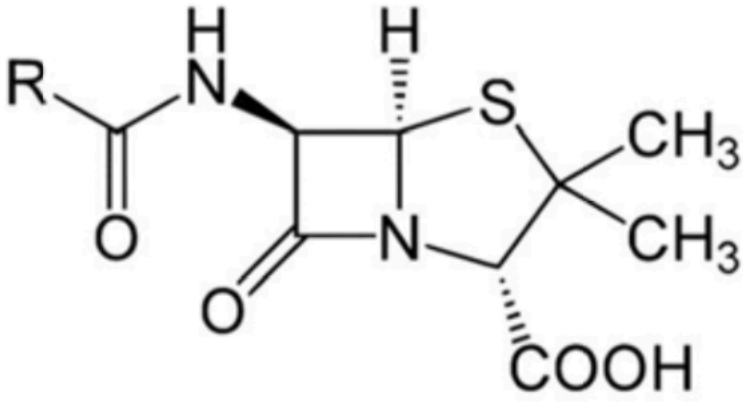
E. Adverse reactions

# Cell Wall Inhibitors

- **B-LACTAM ANTIBIOTICS**

They share a B-lactam ring in their molecular structure which is required for antibacterial activity

- **Penicillin**
- **Cephalosporins**
- **Monobactams**
- **Carbapenem**

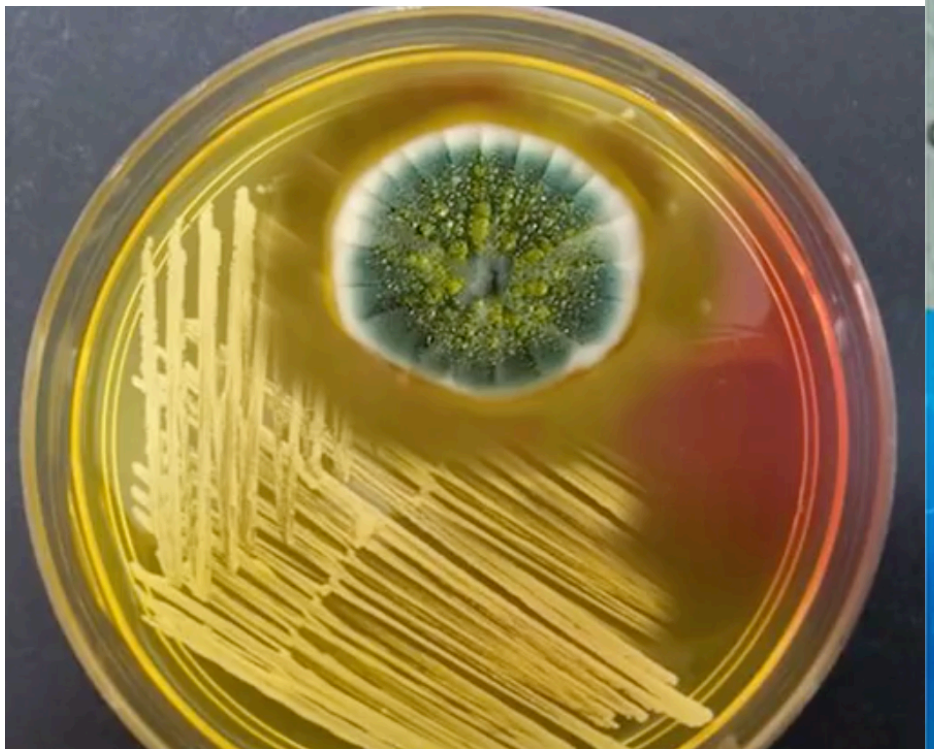



Penicillin core structure, where "R" is the variable group

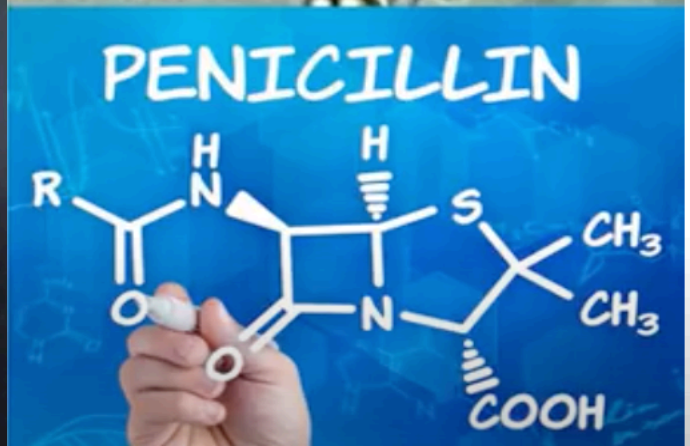




Fleming receives Noble Prize in 1945







# Penicillins

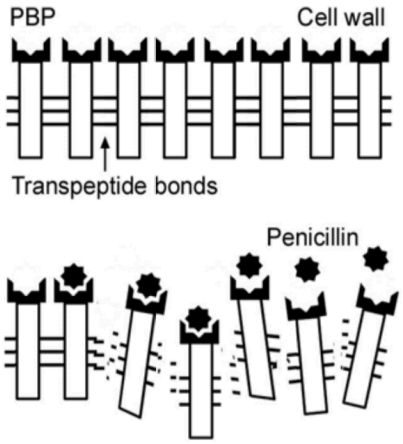
## Antibacterial spectrum:

Its active against rapidly growing micro-organisms including

- **Gram-positive cocci:** e.g. pneumococci and staphylococci. streptococci,
- **Gram-positive bacilli:** anthrax bacillus, C. diphtheria and clostridia.
- **Gram-negative bacilli:** shigella, salmonella, pseudomonas...etc.
- **Spirochetes:** Treponema pallidum.
- **Actinomyces.**

# Mechanism of action:

- Penicillins bind to penicillin binding proteins (PBPs) on the bacterial cell wall and **inhibit transpeptidation** reaction essential for bacterial cell wall synthesis.
- The next step is activation of intracellular autolytic enzymes (autolysins) leading to cell rupture. Thus, the antibacterial effect of penicillin is the result of both inhibition of cell wall synthesis and destruction of existing cell wall by autolysins.
- Penicillins are **bactericidal** especially for Gram-positive bacteria which have thick cell walls.
- The major cause of resistance is the production of **β-lactamase (penicillinase).**



# Resistance

- Survival of bacteria in the presence of β-lactam antibiotics occurs due to the following:

1. **β-Lactamase activity:** This family of enzymes hydrolyzes the cyclic amide bond of the β-lactam ring, which results in loss of bactericidal activity.
2. **Altered PBPs:** Modified PBPs have a lower affinity for β-lactam antibiotics

# Pharmacokinetics

- Food interfere with it's absorption. They must be administered 1hr before or 2 hrs after meals.
- Route of administration of a B-lactam antibiotic is determined by the **stability of the drug to gastric acid** and by **the severity of the infection.**
- Penicillins can penetrate to CSF and ocular fluid only during meningitis.
- They cross placental barrier but they are **not teratogenic**.
- Most penicillins are excreted through the kidney. The renal excretion in proximal tubules could be decreased by co- administration of *probenecid* which prolongs its duration of action.

# Preparations of penicillins:

## 1. Penicillin G and Related drugs:

- Classification:

**A. Injectable Short-acting preparation**, Benzyl penicillin (penicillin-G):

- It is the prototype of all penicillin family.
- Primarily covers gram-positive organisms (**Pneumococci, streptococci, staphylococci (non-penicillinase-producing), Cl. perfringens, C. diphtheria)** with little gram-negative coverage **(Gonococci and meningococci).**
- Others: **Treponema pallidum (syphilis**).
- It has some limitations:

I. It is acid labile and must be given parentrally.

II. It has a short half-life, so frequent injections are required.

III. It is easily inactivated by lactamase enzyme.

IV. It has narrow spectrum.


Penicillin. G S
Sandoz
1,000,000 IU/UI (~0
Only use freshly prepared, clear
que des solutions limpides prép
fois. • Utilicense sólo solucione
Sandoz GmbH, Kundl - Aus
SANDOZ
Penicillin. G Sod./Penici
Sandoz®
5,000,000 IU/UI (~3 g) B.P.
Only use freshly prepared, clear solutions. Only
N'employer que des solutions limpides prépar
A retirer en une seule fois. • Utilicense sólo
recién preparadas. Para una sola toma.
SANDOZ
Sandoz GmbH, A-6250 Kund

## B. Injectable Long-acting preparation

I. **Penicillin G Procaine:** It is a combination of penicillin G with procaine that have longer duration (12-24 hrs). So it is injected just once every day, intramuscularly.

II. **Benzathine Penicillin**: least soluble salt form of penicillin, given intramuscularly, it's absorption lasting from few days to 4 weeks depending on the dose. So it is used in **chemoprophylaxis and in the treatment of syphilis.**




C. **Oral Penicillin V:** it is an acid stable form of penicillin G (given orally).

- Despite widespread use and increase in resistance to many types of bacteria, *penicillin* remains the drug of choice for the treatment of gas gangrene (**Clostridium perfringens**) and syphilis (**Treponema pallidum**).

## 2.Penicillinase or B-Lactamase Resistance Penicillin (Anti-staph Penicillin):

**[Methicillin, cloxacillin, dicloxacillin, flucloxacillin, and nafcillin]**

- They are effective against B-lactamase producing Staphylococci.
- They have low or no activity against other gram-positive and gram-negative organisms.
- The use of these agents is now declining due to increased incidence of methicillin-resistant S. aureus (MRSA).

## 3. Extended spectrum penicillin
## [Ampicillin and amoxicillin]

- They maintain gram-positive coverage with extended coverage of more gram- negative bacteria e.g. E. coli, Salmonella, H. influenza, Proteus, and Shigella (but not Pseudomonas)

- They can be given orally (acid resistant) and by injection (IM and IV).

- They are inactivated by B-lactamase, so they are usually combined with clavulanic acid or sulbactam (B-lactamase inhibitors)

✓ Amoxicillin/clavulanic acid
✓ Ampicillin/sulbactam

- **Ampicillin**

➢ concentrated in bile so it is effective in gall bladder disease and typhoid carrier.

➢ more active on shigella and H. influenza.

- **Amoxicillin**

➢ has better absorption and tissue penetration than ampicillin. So, it has no effect on the gut flora and less G.I. upset.

➢ is more active against Salmonella and Streptococcal fecalis.

➢ can penetrate mucoid and purulent sputum, so it is useful in chronic bronchitis.










Dr. Amr Fouda MD
Antimicrobial Drugs

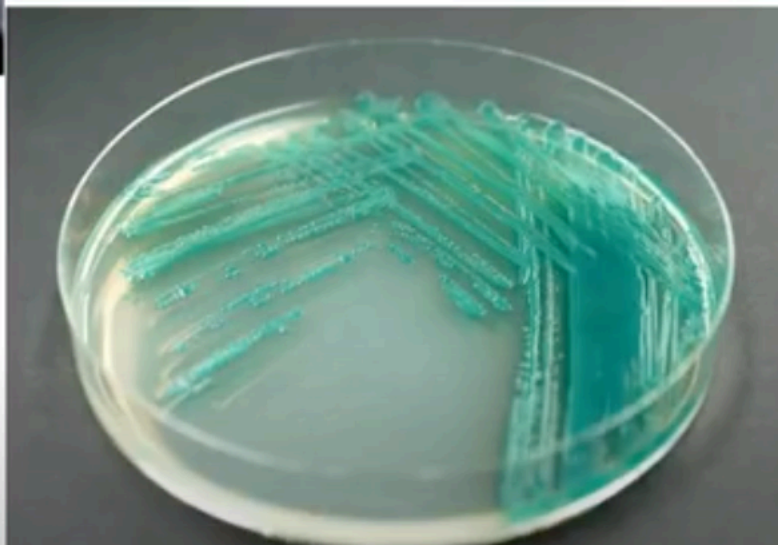
## 4.Antipseudomonal penicillin's

## [Ticarcillin, Azlocillin, and Piperacillin]

- They are broad-spectrum, but should only be used for **pseudomonas infection** and **ampicillin resistant proteus**. Also they have activity against **anaerobic gram negative bacteria e.g. Bacteroids fragilis** (a common pathogen in intra-abdominal sepsis)
- They have synergistic effect when they are used with aminoglycosides.
- They are **inactivated by B-lactamase** so they are combined with **β-lactamase inhibitors** to cover B-lactamase producing bacteria.


1 vial
Piperacillin / Tazobactam
4g/0.5g
Powder for solution for infusion
Each vial contains 4g piperacillin (as sodium salt) and 0.5g tazobactam (as sodium salt)
The sodium content is 218mg per vial
Do not mix or co-administer with any aminoglycoside
Do not reconstitute or dilute with lactated Ringer's (Hartmann's) solution. Contains Penicillin.
Piperacillin and Tazobactam for Injection
4.5 g
4g piperacillin/ 0.5g tazobactam
LATEX FREE STOPPER
BOUCHON SANS LATEX
STERIMAX



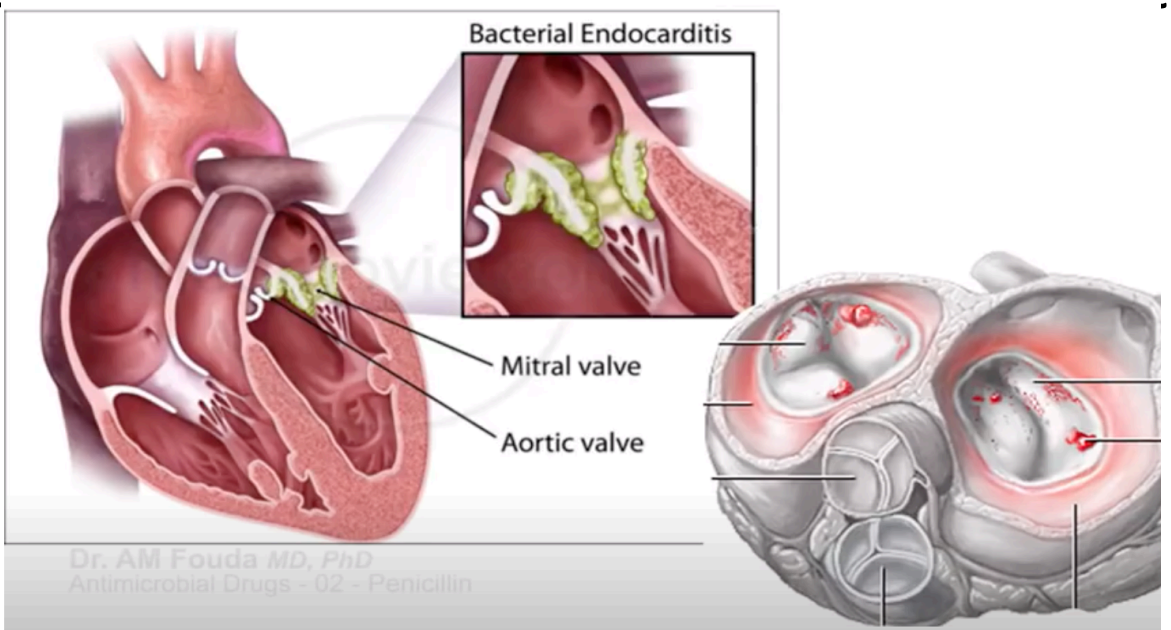


# Therapeutic uses of penicillins:

**A. Treatment:** Penicillins may be used in the treatment of:

1. Streptococcal infections, e.g. wound sepsis, acute throat infections, subacute bacterial endocarditis,.. etc.
2. Staphylococcal infections of skin, mucous membrane and bone.
3. Pneumococcal infections e.g. pneumonia.
4. Syphilis (5 million units single dose) and gonorrhoea.
5. Meningococcal infections: Penicillin diffuses into the CSF only when the meninges are acutely inflamed.
6. Typhoid and paratyphoid fevers: ampicillin and amoxycillin.
7. Pseudomonas infection: (ticarcillin or piperacillin)
8. Actinomycosis, Anthrax and H. influenza infections.
9. Diphtheria, tetanus and gas gangrene. (Penicillin may be used together with the specific antitoxins).

- **B. Prophylaxis:** Penicillins may be used prophylactically in the following conditions:
  1. To prevent recurrence of ***Rheumatic fever***: Benzathine penicillin, 1.2 million units IM once a month.
  2. To prevent ***Subacute bacterial endocarditis*** due to bacteraemia resulting from operative procedures such as dental extraction, tonsillectomy, etc.



# E. Adverse reactions

- **1. Hypersensitivity:** ranging from rashes to angioedema (marked swelling of the lips, tongue, and periorbital area) and anaphylaxis.
- **2. Diarrhea:** Diarrhea is a common problem that is caused by a disruption of the normal balance of intestinal micro-organisms. It occurs to a greater extent with those agents that are incompletely absorbed and have an extended antibacterial spectrum.
- **3. Nephritis:** Penicillins, particularly methicillin.
- **4. Neurotoxicity:** Epileptic patients are particularly at risk due to the ability of penicillins to cause GABAergic inhibition.
- **5. Hematologic toxicities:** Decreased coagulation may be observed with high doses and therapy of greater than 2 weeks.

## Drug interactions:

- Bacteriostatic drugs (e.g. tetracycline, chloramphenicol, erythromycin) interfere with the action of penicillin because penicillin acts by inhibiting cell wall synthesis (i.e. kills rapidly multiplying bacteria) while bacteriostatic drugs arrest the organism and prevents its multiplication.

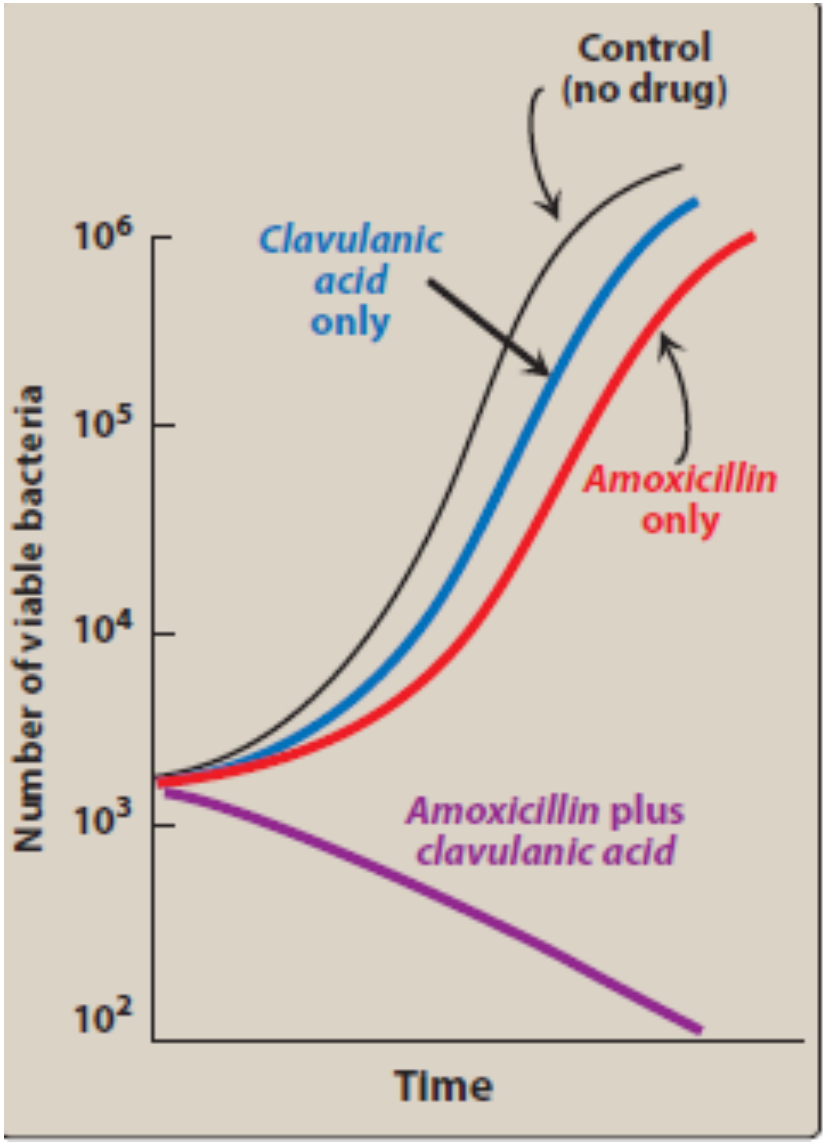
- Antipseudomonal penicillin (acidic drugs, —ve charged) form complex with aminoglycosides (basic drugs, +ve charged) if they are mixed in the same infusion fluid.

# β-LACTAMASE INHIBITORS

- Hydrolysis of the β-lactam ring, either by enzymatic cleavage with a β-lactamase.
- β-Lactamase inhibitors, such as **clavulanic acid, sulbactam, avibactam and tazobactam** by themselves, do not have significant antibacterial activity or cause any significant adverse effects. Instead, they bind to and inactivate β-lactamases, thereby protecting the antibiotics

- The β-lactamase inhibitors are therefore formulated in combination with β- lactamase–sensitive antibiotics. For example, clavulanic acid and amoxicillin on the growth of β-lactamase–producing E. coli.

- [Note: Clavulanic acid alone is nearly devoid of any antibacterial activity.]




The in vitro growth of Escherichia coli in the presence of *amoxicillin*, with and without *clavulanic acid*.

# OTHER β-LACTAM ANTIBIOTICS

## Carbapenems

- Carbapenems are synthetic β-lactam antibiotics ***Imipenem, meropenem, doripenem and ertapenem*** are currently available.
- Imipenem is compounded with cilastatin to protect it from metabolism by renal dehydropeptidase and prevents the formation of the toxic metabolite.

## 1. Antibacterial spectrum:

- ***Imipenem, Meropenem and doripenem*** resists hydrolysis by most β-lactamases,
- This drug plays a role in **empiric therapy** because it is active against β-lactamase–producing gram-positive and gram-negative organisms, anaerobes, and P. aeruginosa .
- Unlike other carbapenems, ertapenem lacks coverage against P. aeruginosa, Enterococcus species and Acinetobacter species.

## 2. Pharmacokinetics:

**Imipenem/cilastatin, meropenem, doripenem and ertapenem are**

- administered IV and penetrate well into body tissues and fluids, including the CSF when the meninges are inflamed. ***Meropenem*** is known to reach therapeutic levels in bacterial meningitis even without inflammation.
- They are excreted by glomerular filtration.

## 3. Adverse effects:

- ***Imipenem/cilastatin*** can cause nausea, vomiting, and diarrhea. High levels of imipenem may provoke seizures.

# Reference

- **Lippincott's Illustrated Reviews Pharmacology 6th edition**